

Mechanical properties and bond characteristics of different fiber reinforced polymer rebars at elevated temperatures



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HIGHLIGHTS

- FRP bars lost up to 55% of their tensile strength at a critical temperature of 325 °C.
- Concrete-FRP bond was significantly reduced upon exposure to high temperatures.
- FRP bars lost up to 81.5% of their bond strength at critical temperature of 325 °C.
- The analytical model, proposed in this study, showed good predictability.

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ABSTRACT

This paper presents the results of an experimental study on the effect of elevated temperatures on the mechanical properties of FRP bars and the bond behavior between FRP bars and concrete. Four types of reinforcement bars namely: basalt fiber reinforced polymer (BFRP), CARBON fiber reinforced polymer (CFRP), glass fiber reinforced polymer (GFRP), and steel bars of 10 mm diameter were used. The results showed that the FRP bars suffered significant reductions in their mechanical properties upon exposure to high temperatures of up to 450 °C at which the GFRP and BFRP melted and lost their total tensile strength capacity. At a critical temperature of 325 °C, the FRP bars lost as high as (55% and 30%) of their tensile strength and elastic modulus, respectively. The percentage reduction in mechanical properties and bond strength was more pronounced in specimens with FRP bars than those with steel bars under elevated temperatures; the percentage reductions in bond strength between concrete and FRP bars reached as high as 81.5% after exposure to 325 °C. Based on the experimental results, an empirical model was proposed to predict the post-heating bond stress-slip relationship between the FRP bar and the surrounding concrete: a good agreement was noticed between the experimental results and the proposed model.

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1. Introduction

During the last twenty years, many studies have been carried out on FRP materials. It was found that these non-corroding composite materials can be applied in concrete construction effectively as alternative to steel reinforcement in three different forms namely externally bonded FRP sheets or plates for repair or strengthening of reinforced concrete structure, main reinforcement in the form of FRP bars, and FRP prestressing tendons in prestressed concrete structures [1,2]. FRPs possess some advantages and disadvantages. Their advantages include high resistance to

corrosion, high strength-to-weight ratios (10–15 times greater than steel), excellent fatigue characteristics (about 3 times that of steel), electromagnetic neutrality and ease as well as speed of application leading to reduce construction costs [1–3]. On the other hand, the disadvantages include high material cost, low ductility with low strain at brittle failure, low shear strength due to poor mechanical properties of the matrix, rapid and severe loss of bond, strength and stiffness at elevated temperatures, poor creep and expansion properties in the case of aramid FRP, and low resistance to alkali environments in the case of glass FRP [2–4].

Upon exposure to fire, FRP composites reach firstly the glass transition temperature T_g at which the resin changes from glassy state to leathery state followed by breaking of the chemical bonds, modular chains of the resin and bonds between the fibers at higher temperature known as the resin decomposition temperature T_d .

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Finally, the ignition and combustion of the composite occur at higher temperatures. The glass transition temperature (T_g) and decomposition temperature (T_d) of the polymer matrix were typically in the range of (65–120 °C) and (300–400 °C), respectively [1,4–8].

FRP bars to concrete bond mechanism depends on the bar's mechanical interlocking and the friction between the surface and the surrounding concrete. Chemical bond does not exist between the FRP bars and concrete contrary to the case with traditional bars. This is referred to the water repellent nature of the resin, used in the manufacture of FRP rebar [9]. Bar diameter, embedment length, FRP bar's modulus of elasticity, concrete strength, concrete cover and bars surface treatment are the main factors that were considered in the evaluation of the bond behavior [10–12]. In order to enhance bonds between FRP bars and concrete, different surface treatments are used such as sand coating, helical wrapping, molded deformation, irregular surface humps by adding excess of resin and braiding of the fibers [9]. Recent findings stated that the bond between FRP bars and concrete was inversely related to bar diameter due to the increase of voids formation at larger contact surfaces [13].

Bond strength between internal FRP reinforcements and concrete depends on the characteristics of resin polymer at the surface of the FRP bar [14,15]. The mechanical properties and microstructure of the resin polymer, especially at the surface layer of the rod, are negatively affected by high temperatures above T_g leading to a loss of bond between FRP and concrete [1].

Due to different transverse coefficient of thermal expansion of FRP bars and concrete, high radial pressure on the surface of the reinforcement is created causing bursting thermal stresses within the concrete. When these thermal stresses reach the concrete tensile strength (f_{ct}), micro-cracks occur leading to weakening of bonds especially when lower concrete covers are used [15–18]. The bond strength between FRP bars and concrete is severely affected even at relatively low elevated temperatures [2,19]. Rapid loss in bond strength, to 60% of room temperature strength, has been noticed at 100 °C. At 200 °C temperature, bond strengths decrease to 10–20% of its room temperature strength. Previous research confirmed that bond is totally lost at a high temperature of 300 °C and greater [1,20].

In this study, the effect of elevated temperature exposure on the mechanical properties of FRP/steel bars and the bond properties between these bars and concrete were investigated experimentally and empirically. Canadian standards, CAN/CSA-S806-02 [21] stipulated that FRP-RC structures fail when the FRP bar loses 50% of its strength at ambient temperature. Due to lack of data on the thermal mechanical properties of wide variety of FRP products, available in the market, the critical temperature for FRP that affects their properties and bond strength with concrete was not precisely defined [22]. Based upon various literature works, the critical temperature can be assumed to range from 250 °C to 500 °C [7,23–27]. The current study showed that the FRP bars had lost as high as (55%, 30% and 81.5%) of their tensile strength, elastic modulus and bond strength with concrete after exposure to 325 °C, respectively.

2. Experimental program

2.1. Materials

2.1.1. Reinforcement bars

Four types of 10 mm diameter reinforcement bars are considered in this work including helically wrapped glass fiber reinforced polymer (GFRP), sand-coated carbon fiber reinforced polymer (CFRP), helically wrapped basalt fiber reinforced polymer (BFRP)

and deformed steel bars (Fig. 1). The properties of the FRP bars used as provided by the manufacturer are presented in Table 1, whereas the properties of the steel bars before and after exposure to 450 °C are presented in Table 2.

2.1.2. Concrete

Normal strength concrete having 28-days-compressive strength of 40 MPa and average measured slump of 90 ± 2 mm was used in preparing the various specimens. Ordinary Portland cement Type-I, crushed granite coarse aggregate of 10 mm maximum size and natural river sand of 2.7 specific gravity and 0.6% moisture content were used. The concrete mixture proportions are presented in Table 3.

2.1.3. High tensile strength epoxy

Epoxy adhesive having the commercial designation CONCRE-SIVE 1441S, and manufactured by BASF-Malaysia, was used to adhere grip steel pipes to the FRP bars free ends before tension and pullout tests. These steel pipes provide a confinement pressure on the bar to prevent its slippage or local failure due to the damage of the bars ends under tension. After 7 days of curing, this epoxy has tensile, compressive and shear strengths of 10, 83 and 35 MPa, respectively, whereas its density and elastic modulus are 1.25 kg/L and 4600 MPa, respectively.

2.1.4. High temperature thermal insulation coating

High-temperature thermal insulation coating (RLHY-12) of 0.03 W/m.K heat conductivity, manufactured by Beijing Ronglihengye Technology Corporation, Ltd – China, was used to insulate the free part of the FRP bars pullout specimens and protect them from direct fire flames during heating.

2.2. Specimens preparation

2.2.1. Preparation of tensile testing specimens

BFRP, CFRP, GFRP and steel bar specimens of 600 mm length and 10 mm diameter were prepared for tensile tests. A total of 18 specimens of each bar type were used. Three of them were tested at ambient temperature and used as controls while the remaining fifteen specimens (in triplicates) were exposed to different elevated temperatures of 125 °C, 250 °C, 325 °C, 375 °C and 450 °C. The specimens were heated up with 10 °C/min heating rate in an electric furnace allowing the hot air to circulate in the furnace until reaching the required target temperature, and then kept at

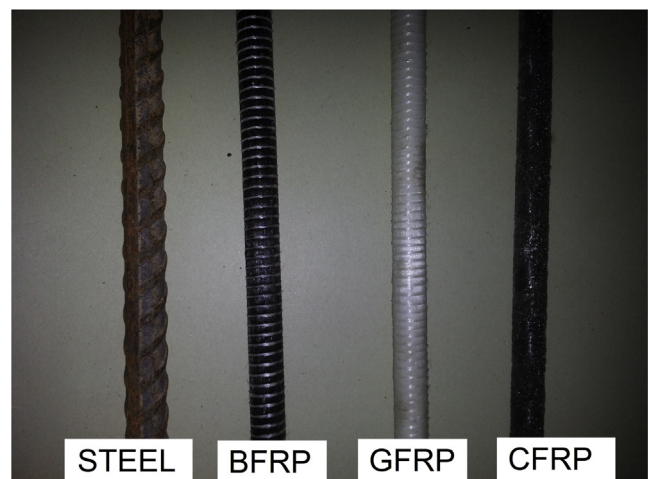


Fig. 1. Different reinforcement bars used.

Table 1

Mechanical properties of different FRP bars used as per manufacturer data sheet.

Type	f_{uf} MPa	ϵ_{uf} %	E_f GPa	$\alpha_T \times 10^{-6}/^\circ\text{C}$	$\alpha_L \times 10^{-6}/^\circ\text{C}$	F %	Type of matrix
CFRP-10 mm	1896	1.6	124	89	-0.9	70%	VER
GFRP-10 mm	800	2.1	40	22	5.4	N.A.	UP
BFRP-10 mm	1000	2.1	65	23	6	N.A.	EPR

f_{uf} , maximum tensile strength; ϵ_{uf} , maximum strain at ultimate strength; E_f , FRP's modulus of elasticity; F%, fiber to matrix volume ratio; α_L , Longitudinal coefficient of thermal expansion; α_T , Transverse coefficient of thermal expansion; VER, Vinyl Ester Resin; UP, Unsaturated Polyester; EPR, Epoxy Polymer Resin; N.A., not applicable.

Table 2

The mechanical properties of reinforcing steel bars used.

Temperature ($^\circ\text{C}$)	YS (MPa)	US (MPa)	EUS (%)
23	570	638	8.48
450	541	622	8.94

YS, Yield Strength; US, Ultimate Strength; EUS, Elongation at Ultimate Strength.

this target temperature for 30 min to ensure a uniform temperature distribution in the FRP specimens [7,28]. After heating, the specimens were left to cool in air and ready for tensile testing. The physical status of the heated GFRP bars to various high temperatures is shown in Fig. 2, as an example. It was noticed that all FRP bars heated up to 450 $^\circ\text{C}$ were totally damaged, hence were not tested for mechanical properties as shown in Fig. 3.

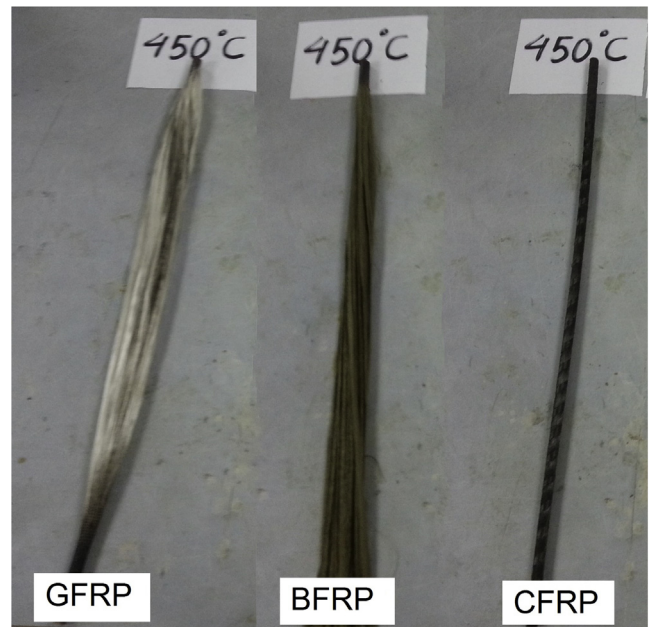
Internally threaded steel pipes with a length, an outer diameter and a thickness of 200, 42 and 4 mm, respectively, were bonded to both ends of the FRP bars using a high tensile epoxy (CONCRETSIVE-1441S) to enable testing of the bars without end failure caused by the testing machine grips. The threaded steel pipes were adhered to the FRP bars after their heat treatment as shown in Fig. 4. The exact elastic modulus of the different specimens were measured using an extensometer.

2.2.2. Preparation of pull-out testing specimens

A total of 60 traditional pullout specimens were prepared using 10-mm-diameter FRP/steel bars embedded vertically in concrete cubes ($150 \times 150 \times 150$ mm) with a bond length of 100 mm. The latter was achieved using bond breakers consisting of PVC pipes of 25 mm length each, as shown in Fig. 5. The de-bonded areas were created to prevent any localized artificial confinement of the bond length due to compressive load on concrete; especially at the loaded end and to ensure a uniform temperature distribution along the entire bar length. The pullout specimens were prepared, cured, and stored at room temperature of 23 $^\circ\text{C}$ for 28 days before testing. Triplicates of pullout specimens with the four types of reinforcement bars embedded in concrete blocks were tested before and after exposed to different temperature levels of 125, 250, 325 and 375 $^\circ\text{C}$ for three hours, and then cooled.

2.3. Mixing, casting, and curing of pull-out specimens

A titling drum mixer of 0.15 m^3 capacity was used in mixing the concrete ingredient following the ASTM Standards [29]. The slump, measured according to ASTM standards [30], was found to be about 90 mm. Pullout specimens, standard cylinders and cubes were cast in three layers, compacted using a vibrating table and then left in their wooden molds for 24 h and covered with wet

**Fig. 2.** GFRP bars before and after exposure to different high temperatures.**Fig. 3.** Damaged FRP bars after exposure to 450 $^\circ\text{C}$.

burlap before being de-molded. After 24 h, all specimens were de-molded and cured for 21 days at 23 $^\circ\text{C}$ using daily sprayed water. The specimens were left in the laboratory air for a week before being heated or tested.

Table 3

Concrete mixture proportions.

Water (kg/m^3)	Cement (kg/m^3)	Coarse aggregate (kg/m^3)	Fine aggregate (kg/m^3)	Target compressive strength (MPa)	Target slump (mm)
207	416	1139	619	40	90



Fig. 4. Samples of FRP bars before and after exposure to different elevated temperatures, ready for tensile strength test.

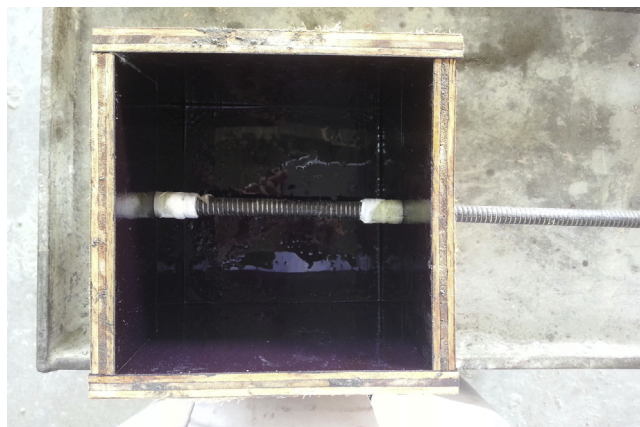


Fig. 5. Pullout test specimen mold with bars embedded to the desired length.

2.4. Heating of pull-out specimens

The pullout specimens were heated to 125 °C and 250 °C in an electric furnace at a rate of 10 °C/min and in a gas furnace at a rate of 100 °C/min when heated at higher temperatures of 325 °C and 375 °C. The target temperature was maintained constant for 3 h to achieve uniform distribution of the heat across the cubic concrete blocks. Thermal insulating coat was also applied on the exposed length of the FRP bars prior to heating in order to protect them from direct gas fire flames that may damage the FRP bars. Figs. 6 and 7 show the heating scheme used for pullout specimens and pullout specimens inside the gas furnace after heating, respectively.

It was noticed that most of pullout specimens with FRP bars heated to 375 °C were damaged, hence were not tested under the pullout bond test. After cooling of the pullout specimens in the laboratory air, threaded steel pipes with a length, an outer diameter and a thickness of 100, 20 and 3 mm, respectively, were bonded to the free end of the FRP bars using high tensile epoxy adhesive (CONCRETE-1441S). This allowed pullout of the FRP bars without end rupture because of the effect of stress concentration at the tensile machine's gripper's location;

2.5. Mechanical testing

2.5.1. Tension test of FRP/steel bars

The tensile tests on FRP/steel bars were conducted at room temperature (23 °C) according to ASTM standards [31] using a

Universal Testing Machine of 1000 kN maximum capacity. The tensile testing were performed using a standard constant head stroke rate of 3.0 mm/min until failure according to ASTM standards [31] recommendations. The tensile force was transmitted from the upper and lower jaws of the testing machine to the bars using epoxy bonded steel pipe at two ends of the tested bars as shown in Fig. 8. Loads versus displacement were acquired automatically, before being analyzed for ultimate tensile stresses or stress-strain diagrams. An extensometer was installed on certain bars representing the different materials (steel/FRP) to enable acquiring the exact strain values as per ASTM standards [31] provisions.

2.5.2. Pull-out test

Using a universal testing machine of 1000 kN maximum capacity, the pullout tests were conducted at room temperature (23 °C) with a displacement-rate control of (1.2 mm/min) loading rate according to ACI standards [32] as shown in Fig. 9. The average bond strength, the bar's elongation and the actual slip were calculated according to the provisions of ACI standards [32].

2.5.3. Compressive and tensile splitting tests of concrete

After 28 days of curing, standard concrete cubes and cylinders were tested for compressive strength, tensile splitting strength and modulus of elasticity before and after exposure to of 500 °C for 90 min. Compression test on standard concrete cubes and cylinders was performed according to BS standards [34] and ASTM standards [33], respectively, whereas the splitting tensile test was performed according to ASTM standards [35]. Strain gauge was fixed to the concrete cylinders to measure the actual compressive strain.

3. Experimental results and discussion

3.1. Post-heating mechanical properties of reinforcement bars

The effect of elevated temperature on the mechanical properties of FRP/steel bars such as the tensile strength, modulus of elasticity and ultimate strain were determined experimentally then the results were compared with predictions by present analytical models such as Nadjai's proposal [36], Wang's Proposal [37] and Eurocode4 [38]. The residual mechanical properties of the heated FRP materials as function of temperature was computed according to Eqs. (1) and (2) by Nadjai et al. [36] and Eq. (3) by Wang et al. [37]. The coefficients k_f and k_E were calculated using coefficient A, B, C, and n proposed by Nadjai and Wang, and graphically presented in Fig. 12.

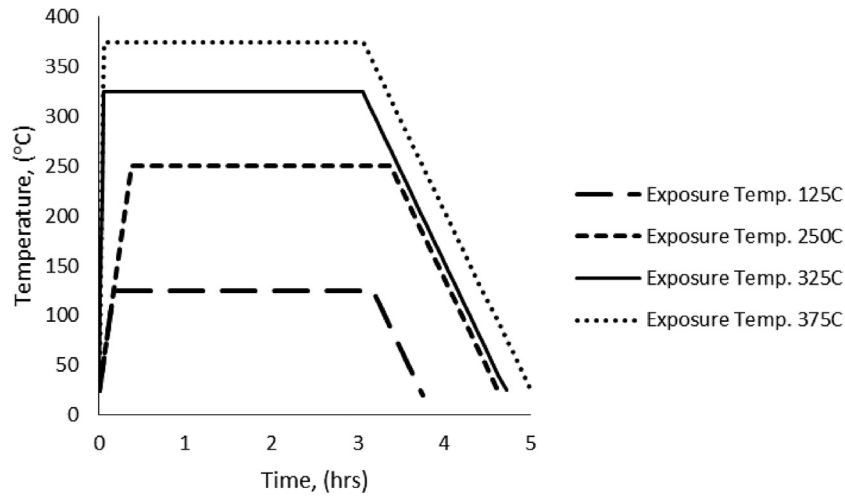


Fig. 6. Heating scheme for pullout specimens.



Fig. 7. Pullout specimens with external thermal insulation after heating in a gas furnace.

$$k_{f,Nadjai} = \text{Residual tensile strength at temperature}(T) = \frac{f_{uf,T}}{f_{uf,20}} \quad (1)$$

$$k_{E,Nadjai} = \text{Residual modulus of elasticity at temperature}(T) = \frac{E_{f,T}}{E_{f,20}} \quad (2)$$

$$k_{f,Wang} = \frac{f_{uf,T}}{f_{uf,20}} = A - \frac{(T - B)^n}{C} \quad (3)$$

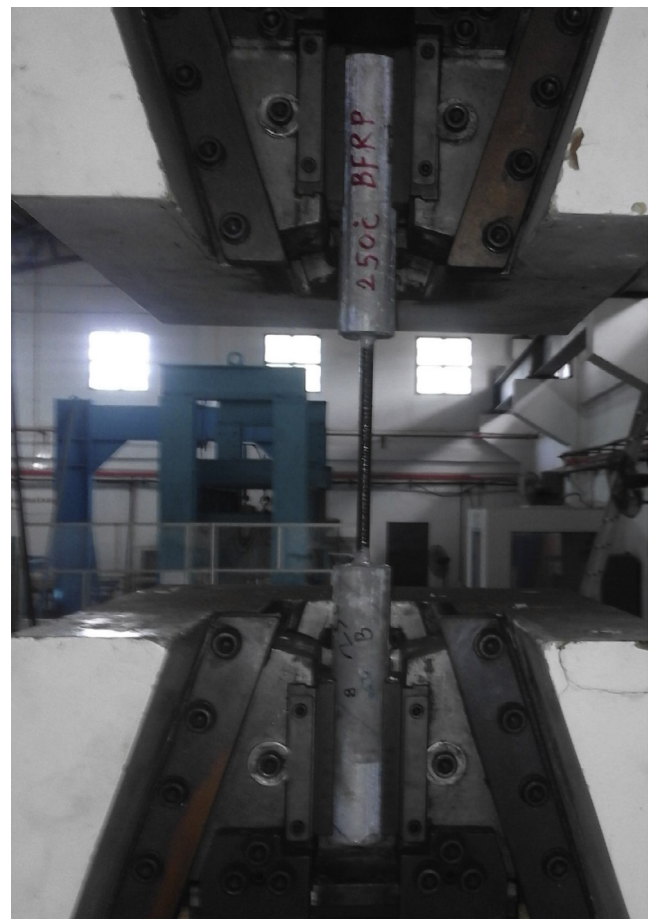


Fig. 8. FRP/steel bars attached to the jaws of the testing machine using epoxy bonded pipes.

Typical stress-strain curves of the different bars before and after exposure to high temperatures in the range of 23 °C–450 °C are shown in Fig. 10 with their average characteristics summarized in Tables 4 and 5; these readings are average of triplicate specimen's readings. The stress-strain relationship of the different FRP bars was almost linear up to failure for temperatures in the range from 23 °C up to 325 °C yet became non-linear at higher temperatures



Fig. 9. Pullout test arrangement.

due to the loss of bonds between fibers resulting from the damage of epoxy resin. The tensile stiffness and elastic modulus were reduced significantly especially prior to failure because the stress distribution among the fibers become non-uniform prior to failure; the pictures of Fig. 11 show the status of fibers upon heating to high temperature.

As indicated, the average ultimate tensile strengths of three specimens of GFRP, BFRP, CFRP and steel bars at ambient

temperatures were 816.5, 940, 1572 and 638.3 MPa, respectively. Their corresponding modulus of elasticity values were 47.14, 65.58, 119.08 and 228.3 GPa, respectively. After exposure to high temperatures, FRP suffered reductions in their tensile strength proportional to exposure temperature whereas exposure to high temperatures up to 450 °C had a minor effect on steel's mechanical properties as compared to that of FRP bars. The critical temperature which caused a significant degradation in the tensile strength of FRP bars in the range of 45–55% was 325 °C. At this critical temperature, the different FRP bars suffered 21–32% loss in their elastic moduli. A higher rate of strength and elastic modulus loss was noticed at 450 °C when the GFRP and BFRP bars melted hence lost their tensile strength capacity totally whereas the CFRP bars retained 10% and 37% of their tensile strength and elastic modulus, respectively.

Upon exposure to (125 °C, 250 °C, 325 °C and 375 °C), GFRP, BFRP and CFRP bars lost (14%, 32%, 44% and 91%), (19%, 30%, 46% and 87%) and (9%, 27%, 55%, 71%) of their ambient temperature strength, respectively. The elastic modulus of FRP is governed by the properties of the fibers hence reduction in elastic modulus with temperature was smaller than that of tensile strength. The residual elastic modulus of GFRP, BFRP and CFRP bars after exposure to (125 °C, 250 °C, 325 °C and 375 °C) was (95%, 87%, 79% and 52%), (96%, 88%, 79% and 53%), (92%, 84%, 68%, and 47%) of their ambient temperature elastic modulus, respectively. Steel bars preserved high portions of their yield strength and elastic modulus upon exposure to high temperatures from 125 °C to 375 °C. Upon exposure to 450 °C, the yield strength and the elastic modulus lost 5.3% and 7.7% of their corresponding room temperature values, respectively.

It was observed that fracture failure of FRP bars exposed to lower elevated temperatures (250 °C and below) happened suddenly within the specimen's middle length. FRP bars heated to 325 °C and greater showed some repeated minor load drops before failure, hence tests were terminated before bars separation, see Fig. 11. Small differences between triplicate readings were noticed at temperatures below 325 °C. The decomposition of the polymer matrix at high temperatures affected the bond between the fibers hence the bar's mechanical properties. Therefore, significant differences between triplicate readings were noticed.

The experimental results were used to assess the accuracy of the available analytical models such as the Nadjai's and Wang's models [36,37]. As shown in Figs. 12 and 13, Nadjai's model tended to overestimates the predicted reductions in tensile strength and elastic modulus of GFRP and BFRP bars yet showed a good agreement with those of CFRP bars. On the other hand, Wang's model tended to overestimate the reductions in tensile strength and elastic modulus of FRP bars at temperatures below 325 °C yet significantly underestimated those reductions especially at higher temperatures. The reduction in tensile strength and elastic modulus of FRP bars is almost linear from ambient temperature up to 325 °C.

The normalized yield tensile strength and normalized elastic modulus of steel bars were compared with reduction factors presented in Section 3.2.3 of Eurocode4 [38]. As shown in Fig. 14, the Eurocode equations assume that the yield strength is not affected by temperature less than 300 °C, disagreeing with the experimental results. Furthermore, the Eurocode equations tend to overestimate the reduction in the yield strength at temperatures greater than 300 °C compared with the experimental results. According to Eurocode equations, the elastic modulus is not affected by temperature of less than 100 °C whereas reductions in the range of 13%–60% is predicted with heating from 200 °C to 500 °C, respectively. Eurocode equations tend to overestimate the reduction in elastic modulus at temperatures greater than 150 °C compared with the experimental results.

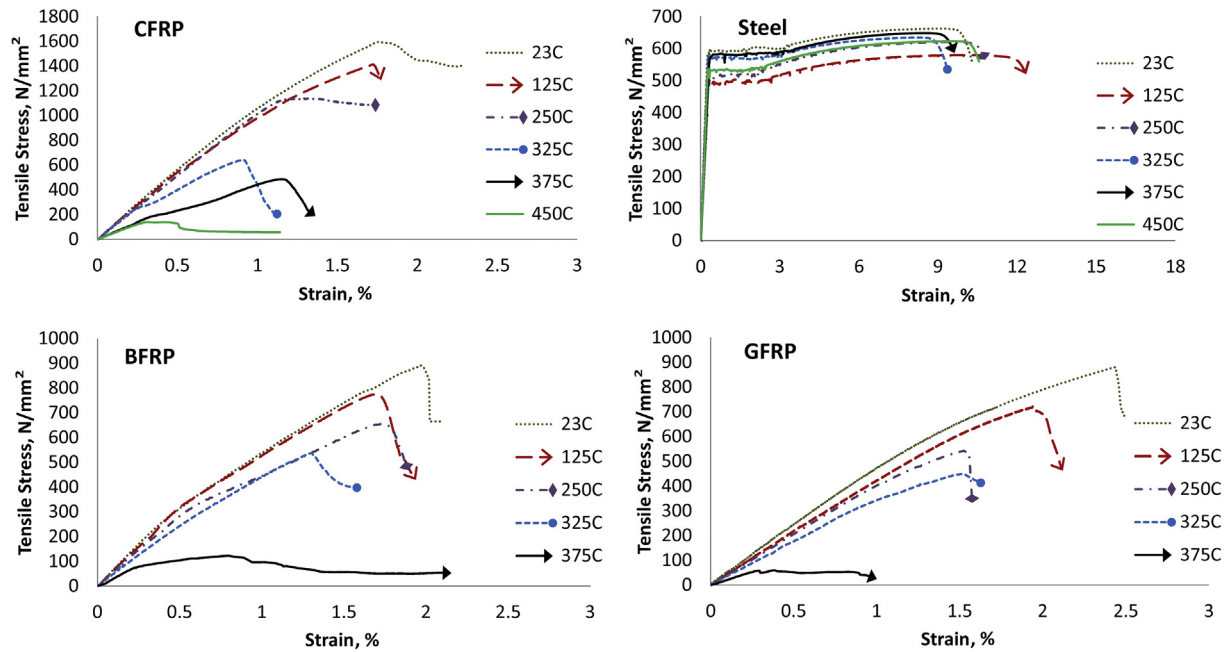


Fig. 10. Typical stress-strain curves of different bars at different temperatures.



Fig. 11. Failure of heated FRP bar in tensile testing.

3.2. Residual mechanical properties of concrete

Average compressive and tensile splitting strengths of concrete cylinders specimens, tested before and after exposure to a high temperature of 500 °C for 90 min, are presented in Table 6.

According to Eurocode [39], the reduced concrete compressive strength after heating to 500 °C is approximately 60.5% of its unheated value, which is close to that of Zhang et al. [40] at 65%. In the present work, the experimental average cylinder compressive strength (f_c') was found to be 34.0 MPa before exposure to heating but was reduced to 25.2 MPa after

heating to 500 °C at a residual value of 74%. The corresponding average cube compressive strength (f_{cu}) was 41.0 MPa as compared to 31.3 MPa after heating. The corresponding average strain at ultimate stress increased from 0.00260 before to 0.006489 after heating.

According to Eurocode [39] the theoretical residual splitting tensile strength of concrete after exposure to 500 °C is 20% as compared to 35% and 45% calculated by Zhang et al. [40] and Noumowe et al. [41], respectively. The experimental average splitting tensile strength was 3.265 MPa before as compared to 1.706 MPa after heating with a residual value of 52%.

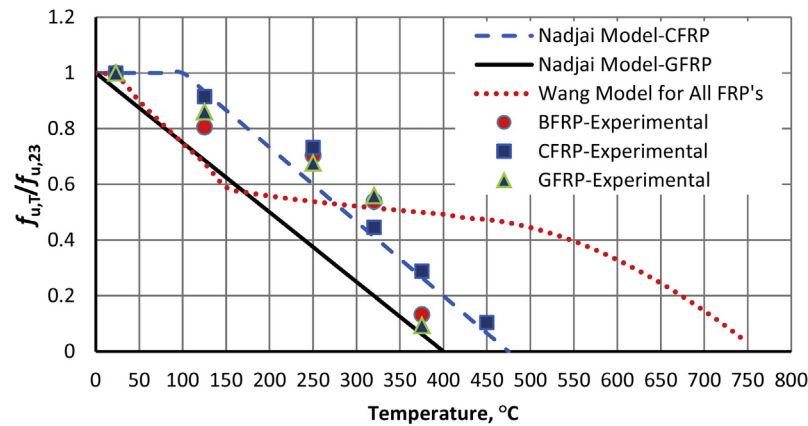


Fig. 12. Residual tensile strength for FRP bars under elevated temperatures.

Table 4

Average characteristics of stress-strain diagram for different FRP bars at different temperatures.

FRP Type	Temp.	f_u MPa	ϵ_{ult} % mm/mm	E_f GPa	$f_{u,T}/f_u$	$E_{f,T}/E_f$
GFRP	23 °C	816.49	2.22	47.14	1.00	1.00
	125 °C	703.29	1.87	44.87	0.86	0.95
	250 °C	552.25	1.59	40.91	0.68	0.87
	325 °C	452.97	1.41	37.40	0.55	0.79
	375 °C	76.74	0.40	24.50	0.09	0.52
BFRP	23 °C	939.99	2.14	65.58	1.0	1.0
	125 °C	757.62	1.61	63.02	0.81	0.96
	250 °C	662.37	1.48	57.74	0.70	0.88
	325 °C	514.28	1.30	51.89	0.55	0.79
	375 °C	122.28	0.791	30.73	0.13	0.47
CFRP	23 °C	1572.07	1.70	119.08	1.0	1.0
	125 °C	1438.33	1.67	109.76	0.91	0.92
	250 °C	1152.30	1.40	100.56	0.73	0.84
	325 °C	700.02	0.95	81.10	0.45	0.68
	375 °C	454.41	1.15	56.23	0.29	0.47
	450 °C	164.27	0.48	43.81	0.10	0.37

f_u : ultimate tensile strength, ϵ_{ult} : strain at ultimate tensile strength, E_f : tensile chord modulus of elasticity.

Table 5

Steel bars-tensile tests results.

Temp.	F_y MPa	F_u MPa	ϵ_{ult} % mm/mm	E_L % mm/mm	E_s GPa	$F_{y,T}/F_{y,23^\circ C}$	$f_{u,T}/f_{u,23^\circ C}$	$E_{s,T}/E_{s,23^\circ C}$	$E_{L,T}/E_{L,23^\circ C}$
23 °C	571.46	638.24	8.48	10.54	228.33	100.0%	100.0%	100.0%	100.0%
125 °C	513.01	598.70	9.73	11.72	203.92	89.8%	93.8%	89.3%	111.2%
250 °C	524.90	619.74	9.34	11.16	207.74	91.9%	97.1%	91.0%	105.9%
325 °C	547.32	628.26	9.23	10.97	209.21	95.8%	98.4%	91.6%	104.1%
375 °C	554.37	638.97	9.07	10.72	213.14	97.0%	100.1%	93.3%	101.7%
450 °C	541.12	622.45	8.94	10.30	210.72	94.7%	97.5%	92.3%	97.8%

F_y : yield strength, F_u : ultimate tensile strength, ϵ_{ult} : strain at ultimate tensile strength, E_L : Elongation at Rupture, E_s : Modulus of Elasticity.

3.3. Bond strength test results

3.3.1. Bond between FRP/steel bars and concrete at ambient temperature

Typical bond slip curves representing the different specimens and exposure temperatures were prepared to study the bond performance of each type of FRP bars with concrete under various exposure temperatures. Fig. 15 demonstrates bond versus slip between concrete and GFRP, BFRP, CFRP and steel bars for pullout specimens maintained at ambient temperature before testing. As shown, the bond slip curves consist mainly of three portions; ascending linear followed by nonlinear portions then irregular descending portion of the curve. The descending part showed non-linear softening behavior reflecting the degradation of the bond.

The behavior obtained in Fig. 15 is in agreement with previous research which concluded that the bond strength between reinforcing bars and concrete is influenced by the bar's surface characteristics, elastic modulus; the surface of all the FRP bars were damaged when observed after the pullout testing.

As expected the steel bars attained the highest bond with concrete at 11.31 MPa followed, in sequence, by the CFRP, BFRP, and GFRP at 8.338, 2.628 and 2.014 MPa, respectively. On the other hand, the CFRP bars showed the highest bond stiffness yet the lowest slip at failure among all the FRP bars due to better surface compared with the GFRP and BFRP bars.

3.3.2. Post-heating bond behavior between FRP/steel bars and concrete

Since the FRP bars have high transverse coefficient of thermal expansion as compared to concrete, hence the surrounding con-

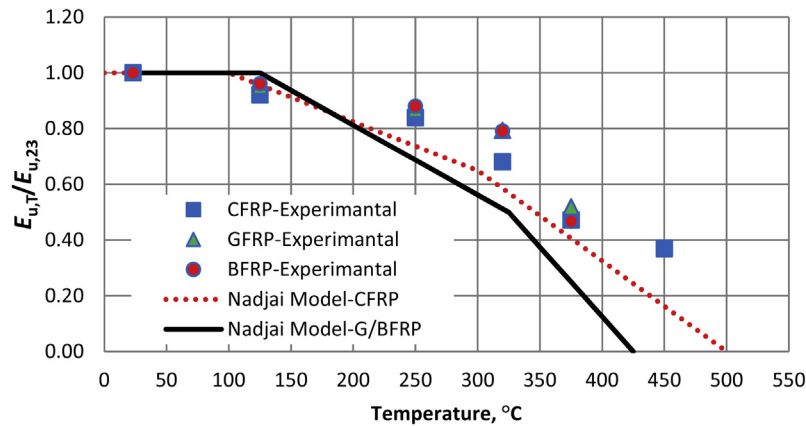


Fig. 13. Residual elastic modulus for FRP bars under elevated temperatures.

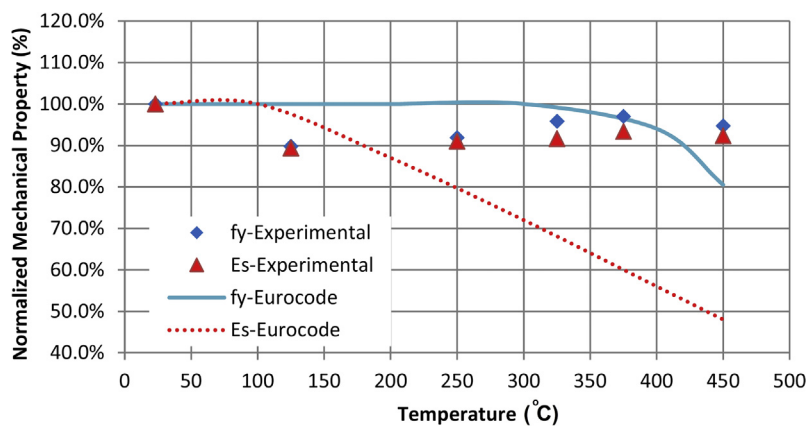


Fig. 14. Normalized tensile strength and elastic modulus of steel bars.

Table 6

Compressive and tensile strength of concrete at 23 °C and 500 °C.

Temp. (°C)	P_c (kN)	f'_c (MPa)	ϵ_c (%)	P_T (kN)	f_t (MPa)
23	266.9	33.98	0.2635	102.58	3.265
500	198.2	25.24	0.6489	53.60	1.706

P_c : maximum measured compression load, f'_c : concrete cylinder compressive strength at 28 days, ϵ_c : Strain at f'_c , P_T : maximum measured tensile load, f_t : splitting tensile strength.

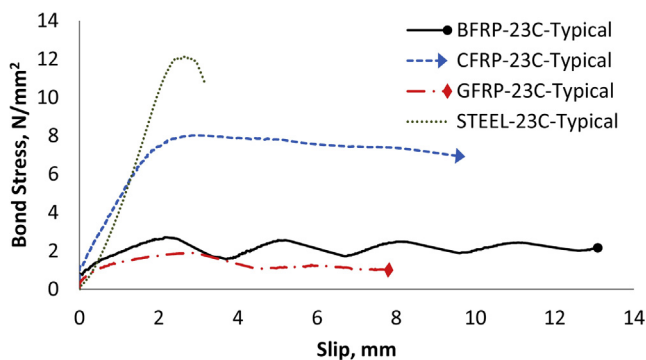


Fig. 15. Typical bond-slip curves of FRP and steel bars at ambient temperature.

crete would be bursted at a relatively low exposure temperature; reducing bond strength between the bars and concrete. The reduction in bond is aggravated further due to the undesirable change in the mechanical properties of the surface polymer resin upon heat-

ing. Accordingly, two types of failure modes were noticed. The first was represented in pullout of the bar due to concrete splitting with radial cracking along the embedded length of the bar. The second failure mode occurred when the bond strength is lower than the splitting tensile strength of concrete: pullout of the bars occurs due to the rather weak abrasion of the exterior surface of the bars. The first mode is expected to be predominant in pullout specimens subjected to temperatures lower than the glass transition temperature (T_g) of the FRP bar; whereas the second is most probably to happen in specimens subjected to temperatures higher than the glass transition temperature (T_g) [8,14–18].

3.3.2.1. Bond stress versus slip relationship. The above discussion explained the clear degradation in the bond characteristics of the FRP bars upon exposure to varying high exposure temperatures. Fig. 16 demonstrates that the changes in the drawn bond stress versus slip curves were affected by both the type of the bar and exposure temperature. In general the curves of the bond stress versus slip pertaining to GFRP and BFRP experience limited linearity

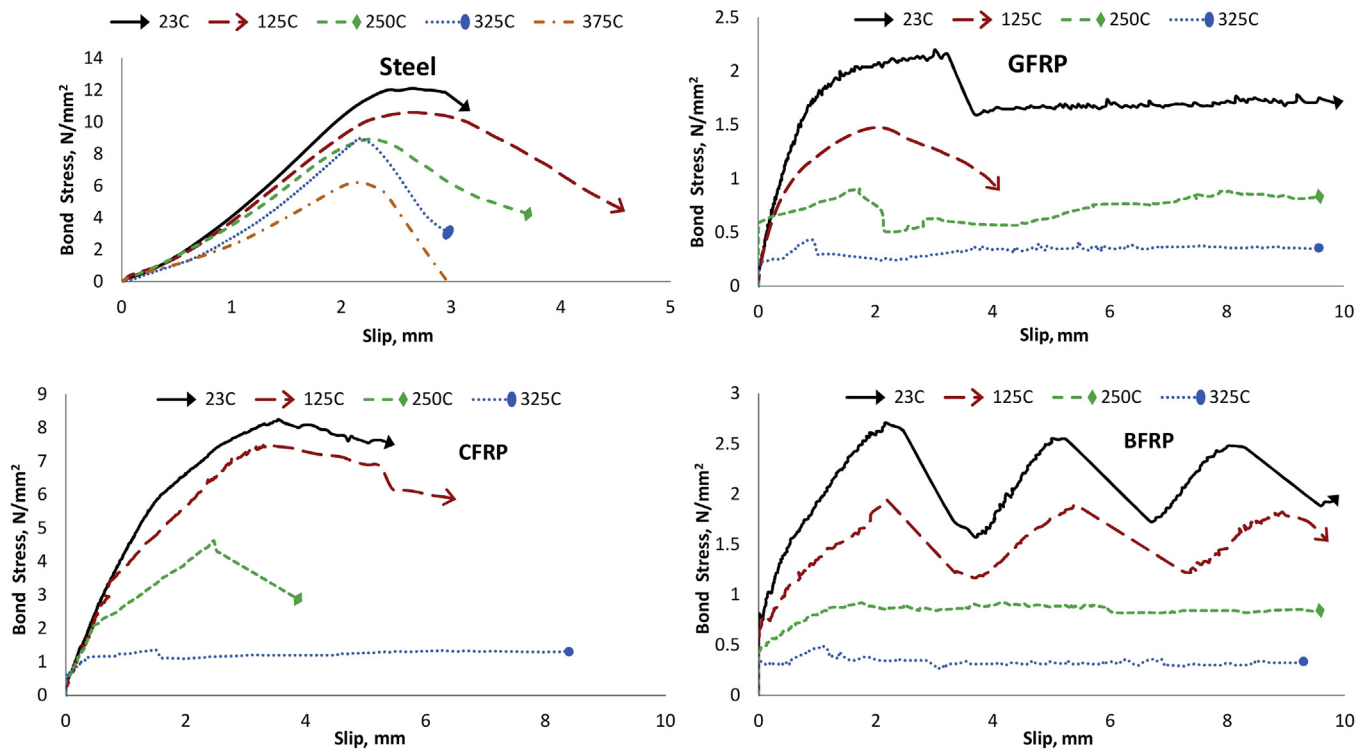


Fig. 16. Typical bond stress-slip curves of different pullout specimens subjected to different temperatures.

with noticeable softening at temperatures greater than 23 °C. The slope of the ascending portion of the bond stress versus strain curves was reduced by exposure temperature, accompanied with reduction in maximum bond stress. As expected, the CFRP and steel bars maintained high bond with concrete up to 250 °C and 325 °C, respectively, beyond which the corresponding bond stress versus slip curves tend to show noticeable softening, as shown in Fig. 16.

3.3.2.2. Characteristics of bond stress versus slip curves. The bond characteristics of different pullout specimens are summarized in Table 7; representing the average of three specimens. As indicated

earlier, the different characteristics were affected by both the type of reinforcement bar as well as the exposure temperature.

Among the pullout specimens of FRP bars, the CFRP ones showed the highest residual bond strength with concrete when exposed to temperatures of (125, 250 and 325 °C) at values of (84%, 56.4% and 18.5%) followed by those of the BFRP and GFRP at (78.2%, 39.1%, 21.1) and (72.6%, 49.8%, 20.8), respectively. The corresponding slip values at maximum bond stress were the highest for pullout specimens with the CFRP bars followed, in sequence, by the BFRP and GFRP at (3.028, 2.224, and 1.607 mm), (2.158, 1.962, and 0.909 mm), and (2.060, 1.892, and 1.048 mm), respectively.

Table 7

Characteristics of bond stress versus strain curves for pullout specimens with different bars and exposure temperatures.

Bar Type	Temp. (°C)	F _u (kN)	SAF (mm)	E _L (mm)	UBS (MPa)	SABS (mm)	RUBS/UBS
GFRP	23	6.328	4.022	1.282	2.014	2.740	100%
	125	4.593	3.038	0.977	1.462	2.060	72.6%
	250	3.150	2.627	0.735	1.003	1.892	49.8%
	325	1.316	1.384	0.336	0.419	1.048	20.8%
BFRP	23	8.256	3.635	1.197	2.628	2.438	100%
	125	6.453	3.136	0.978	2.054	2.158	78.2%
	250	3.227	2.496	0.534	1.027	1.962	39.1%
	325	1.741	1.319	0.320	0.554	0.999	21.1%
CFRP	23	26.198	5.380	2.101	8.339	3.279	100%
	125	22.007	4.943	1.915	7.005	3.028	84%
	250	14.765	3.626	1.402	4.700	2.224	56.4%
	325	4.844	2.177	0.570	1.542	1.607	18.5%
	375*	2.719	2.729	0.462	0.865	2.267	10.4%
Steel	23	35.54	4.45	0.99	11.31	2.85	100%
	125	33.59	3.79	1.05	10.69	2.74	94.5%
	250	28.45	3.16	0.87	9.06	2.28	80.0%
	325	25.87	2.71	0.79	8.24	1.93	72.8%
	375	24.59	3.35	0.73	7.83	2.62	69.2%

F_u: maximum measured force, SAF: stroke at maximum force, E_L: bar elongation, UBS: bond strength, SABS: slip at maximum bond strength, RUBS: residual ultimate bond strength, EC: end cap

* One specimen.

As expected, the steel bars showed the best performance among all bars and sustained higher temperature exposures while still maintaining satisfactory bond. The residual bond strength of pull-out specimens with steel bars were (94.5%, 80%, 72.8% and 69.2%) at exposure temperatures of (125, 250, 325 and 375 °C), respectively. The corresponding slip at maximum stress showed close values to those of the CFRP bars at values of (2.74, 2.28, 1.93 and 2.62 mm), respectively, which were lower than those of the GFRP and BFRP due to stiffer bond behavior of specimens with steel bars.

3.3.3. Failure modes of pull-out specimens

Pullout specimens with GFRP and BFRP failed due to the damage and shearing-off of the surface matrix layer from the core of the slipped bars as shown in Fig. 17a. After breaking the pullout specimens, the residue of the rebar's surface polymers was seen on the concrete whereas no residue of concrete was seen on the bar's surface. The pullout specimens, tested after exposure to 325 °C, showed a high slip value at a low load level indicating that the bond between the FRP bars and concrete was almost damaged. Pullout specimens with the GFRP and BFRP, which were heated to 375 °C, were totally damaged, hence were not tested.

Pullout failure of unheated specimens with the CFRP bars was accompanied by cone failure as shown in Fig. 17b. Upon heating, the sand-coating layer of the carbon bar peeled off when shear strength at the interface between the coating layer and bar core was less than the shear strength of the concrete cover; causing a damage in the bond between the carbon bars and surrounding concrete.

A sudden splitting concrete failure was noticed in the pullout specimens with steel bars accompanied with the formation of 2–4 splitting cracks along the embedded length where the steel bar ribs remains un-damaged. This splitting failure occurred because the bond strength exceeded the splitting tensile strength of concrete which was clearer in the cases of heated specimens due to effect of high temperatures in reducing the splitting tensile strength of concrete. After reaching the maximum bond stress for pullout specimens with FRP/steel bars, the slip continued with gradual reduction in the bond stress.

4. FRP to concrete bond empirical modelling

Different models were analytically developed to predict the ascending branch of the bond stress versus slip diagram; among which was the model proposed by Ciampi et al. [42]. Ciampi et al. [42] provided the following general equation for bond-slip relation of any reinforcing bars type:

$$\tau_{(s)} = \tau_{max} \left(\frac{s}{s_{max}} \right)^{\beta} \quad (4)$$

Where $\tau_{(s)}$ is the local bond stress at local slip (s); τ_{max} and s_{max} are the maximum bond stress and its corresponding slip, respectively; and β is a parameter derived from curve fitting process.

Eq. (4) was applied to the present results after a new parameter (φ_T), which accounts for the temperature effect, was introduced. The modified form of Ciampi's equation is written as follows:

$$\tau_{(s,T)} = \varphi_T \times \tau_{max,23^\circ} \times (s)^{\beta_T} \quad (5)$$

where $\tau_{(s,T)}$ is the local bond stress of a FRP specimen that was exposed to temperature (T) at corresponding local slip (s); $\tau_{max,23^\circ}$ is the bond strength of the FRP bar measured at normal temperature (23 °C); (s) is the local slip which must not exceed the expected maximum slip value ($s_{max,T}$) for each exposure temperature T .

The present results were used to develop a new equation for maximum slip value of FRP specimen exposed to Temperature (T). The equation is given as:

$$s_{max,T} = s_{max,23} \times \left(1 - \frac{T^{2.287}}{10^6} \right) \quad (6)$$

α_T and β_T are parameters that depend on the exposure temperature (T) and given by the following equations:

$$\varphi_T = A \times |0.672 - B - 5.285T \times 10^{-4} - 2.983T^2 \times 10^{-6}|^C \quad (7)$$

$$\beta_T = D \left[0.507 - \left(\frac{T}{1314} \right)^{0.856} \right]^E + F \quad (8)$$

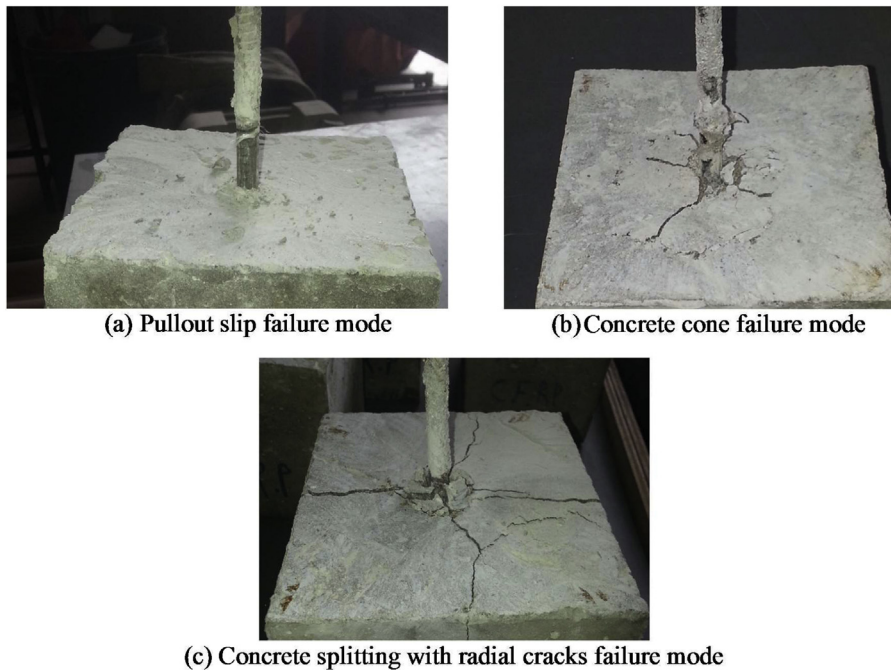


Fig. 17. Different failure modes of pullout tests.

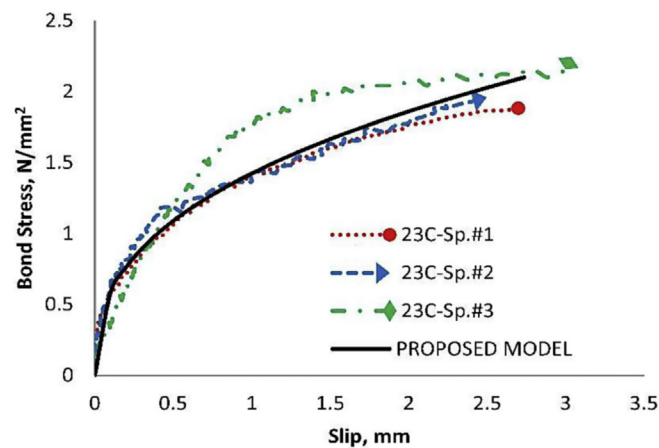
Table 8Regression constants for predicting α_T and β_T (Eqs. (7) and (8)).

FRP type	A	B	C	D	E	F
BFRP	1.0	0.0	1.0	1.0	1.0	0.0
GFRP	1.054	0.115	0.656	3.135	3.590	0.169
CFRP	0.564	0.184	0.186	1.047	0.582	0.0

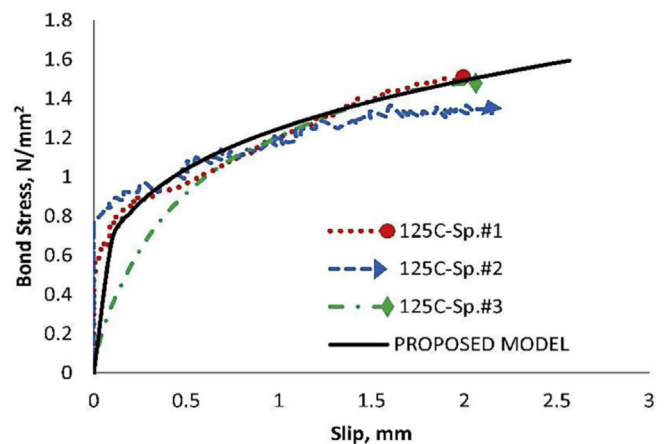
Table 9

Experimental and analytical FRP's maximum bond stress and corresponding slip at different temperatures.

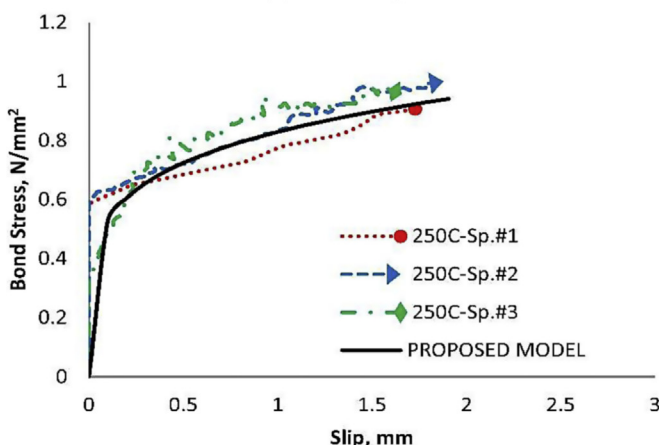
FRP type	Temp. °C	BS _{exp} MPa	S _{exp} mm	BS _{emp} MPa	S _{emp} mm	Error in BS. 100%	Error in S. 100%
GFRP	23	2.014	2.740	2.100	2.736	4.2%	0.2%
	125	1.462	2.060	1.594	2.568	9.0%	24.7%
	250	1.003	1.892	0.940	1.904	6.2%	0.7%
	325	0.411	1.048	0.385	1.218	6.3%	16.2%
BFRP	23	2.628	2.438	2.642	2.435	0.5%	0.1%
	125	2.054	2.158	2.002	2.286	2.5%	5.9%
	250	1.027	1.942	1.068	1.695	4.0%	12.7%
	325	0.554	0.999	0.495	1.084	10.7%	8.5%
CFRP	23	8.339	3.279	9.163	3.274	9.9%	0.1%
	125	7.005	3.028	7.604	3.074	8.5%	1.5%
	250	4.700	2.224	5.035	2.279	7.1%	2.5%
	325	1.542	1.607	1.564	1.457	1.4%	9.3%
Average error						5.9%	6.9%

BS_{exp}: experimental bond strength, S_{exp}: experimental slip at maximum bond stress, BS_{emp}: empirical bond strength, S_{emp}: empirical slip at maximum bond stress.

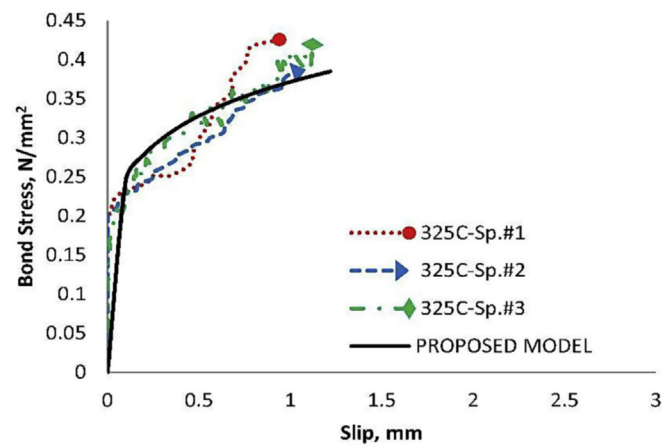
(a) GFRP @ 23°C



(b) GFRP @ 125°C



(c) GFRP @ 250°C



(d) GFRP @ 325°C

Fig. 18. Experimental and empirical ascending portion of bond-slip curves for pullout specimens with GFRP bars exposed to different temperatures.

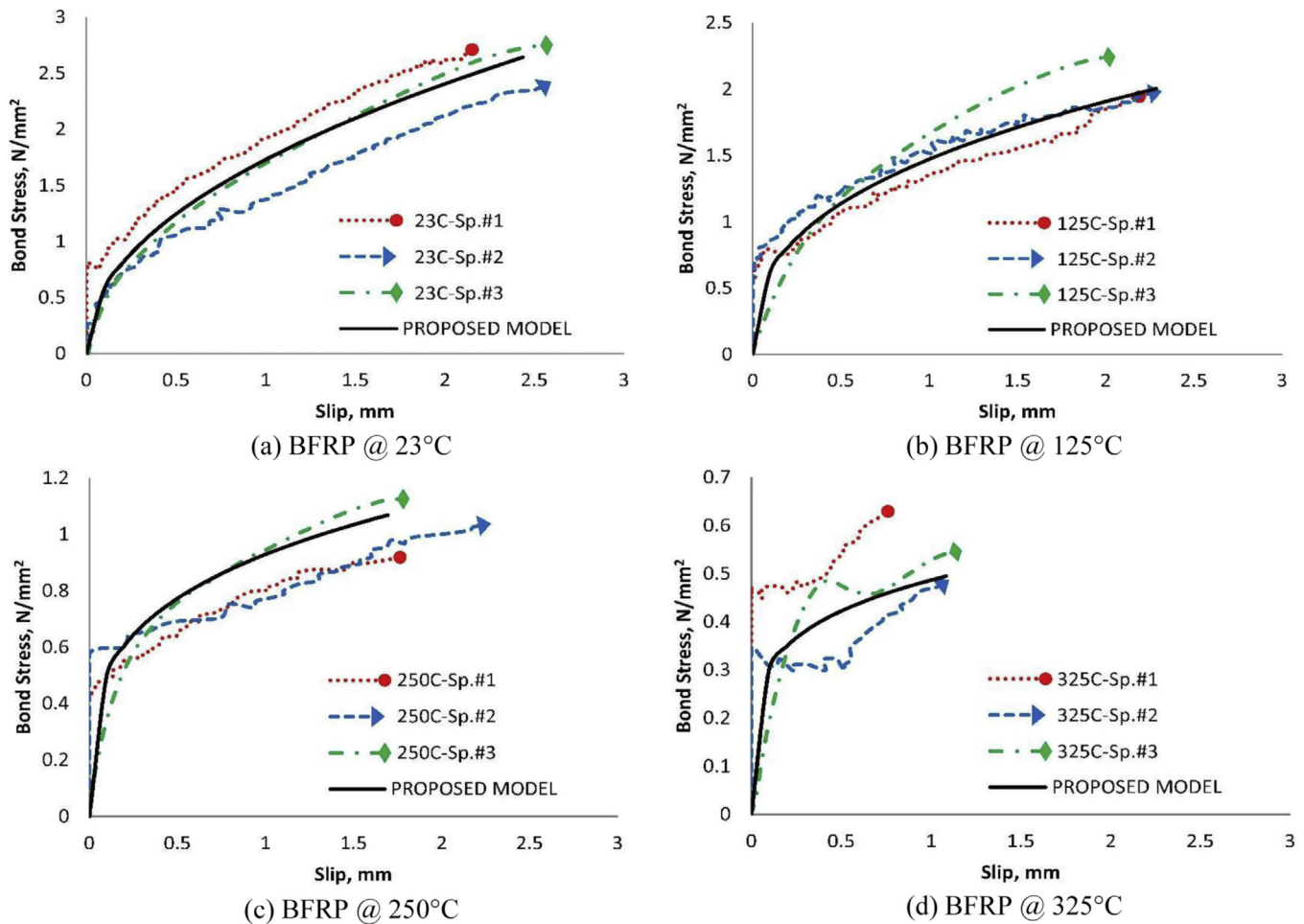


Fig. 19. Experimental and empirical ascending portion of bond-slip curves for pullout specimens with BFRP bars exposed to different temperatures.

where A, B, C, D, E & F are constants estimated based upon curve fitting process of each type of FRP bars; Results are summarized in Table 8.

Based upon Eqs. (2) through (5) and constants of Table 8, predictions of the maximum bond stress were made. Results representing the different FRP bars and exposure temperature were then obtained and summarized in Table 9, before data were presented graphically in Figs. 18 through 20 along with the triplicate readings obtained experimentally. Sample of calculations was presented in Appendix A.

Comparison between the fit of the present model with the experimental data revealed a very good fit in terms of the experimental data scattering around the empirical curve or their trend versus that of the model, Figs. 18 through 20. Furthermore, the predictions of the model of post-heating bond strength at various temperatures for the pullout specimens with FRP bars were consistent with the behavior observed in the laboratory.

5. Conclusions

The impact of heating on tensile strength and elastic modulus of steel bars was minor compared with FRP bars. Significant reductions in the mechanical properties of FRP bars were found upon exposure to temperature of up to 450 °C at which the GFRP and BFRP bars melted and totally lost their tensile strength capacity. The reduction in the tensile strength and elastic modulus of the

FRP bars is almost linear up to a critical temperature of 325 °C at which (45–55%) and (20–30%) reduction percent were noticed, respectively. Concrete lost (25–35%) and 50% of its compressive and splitting tensile strengths, respectively upon exposure to 500 °C.

As expected, the steel bars attained the highest bond strength and stiffness whereas, the CFRP showed the highest bond strength with concrete among the FRP bars because of their better surface characteristics, followed in sequence, by the GFRP and BFRP. The percentage reduction in bond strength and stiffness was more pronounced in pullout specimens with FRP bars than those with steel bars under elevated temperatures; the percentage reductions in bond strength between concrete and BFRP, GFRP, CFRP and steel bars after exposure to (125, 250 and 325 °C) were (21.8%, 60.9%, and 79.9%), (27.4%, 51.2%, and 79.2%), (16%, 43.6%, and 81.5%) and (5.5%, 20% and 27.2%), respectively. Therefore, the critical temperature at which the FRP bars had lost most of their bond with concrete was 325 °C. Three types of failure modes were noticed in the pullout specimens. Pullout failure occurred in specimens with GFRP and BFRP bars, whereas concrete splitting failure was observed in those with steel bars, especially upon exposure to 325 °C and higher. In the control specimens with CFRP bars, the concrete showed cone failure. The empirical model, proposed to predict the post-heating bond stress-slip relationship between FRP bar and surrounding concrete, was in good agreement with the experimental results.

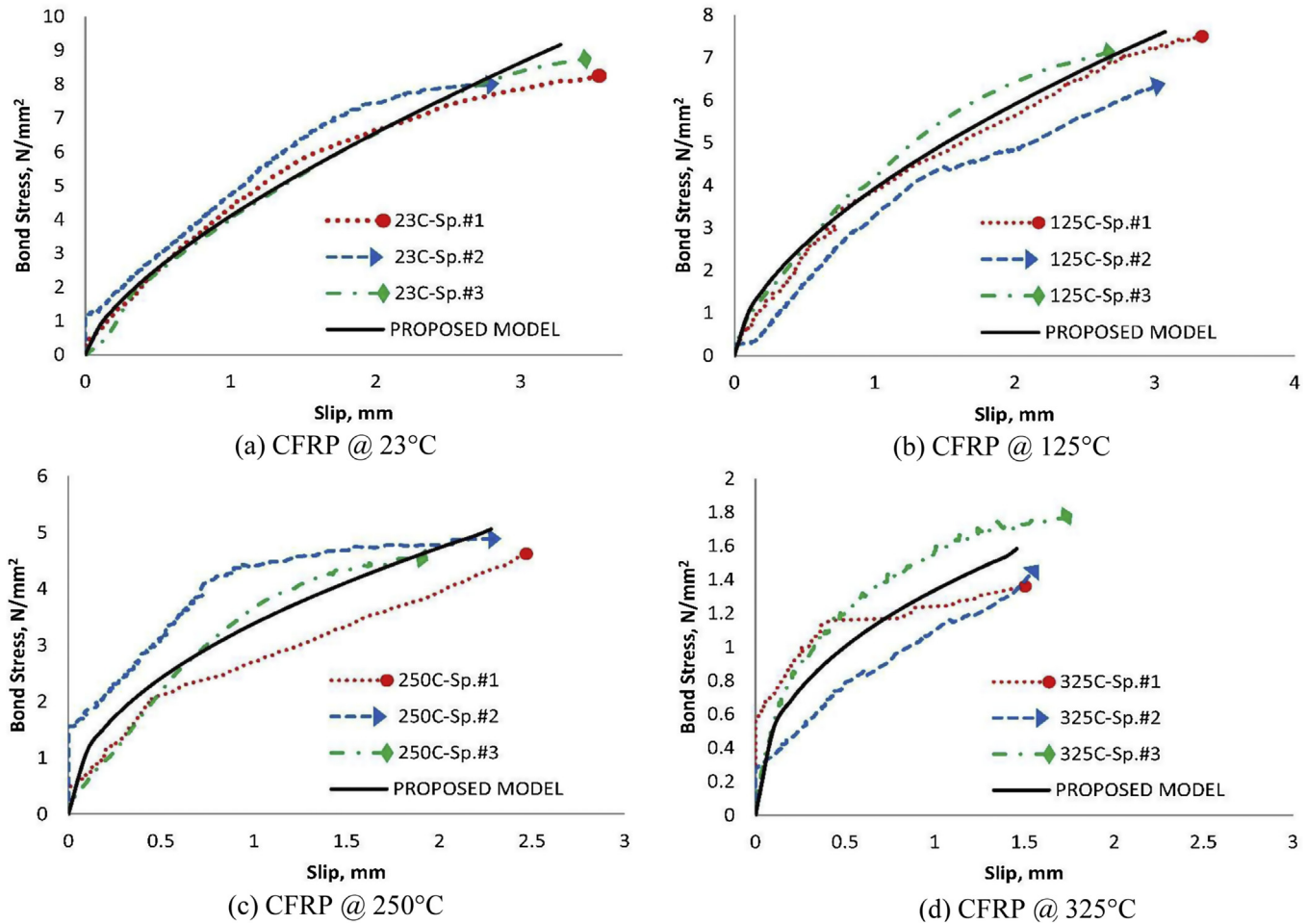


Fig. 20. Experimental and empirical ascending portion of bond-slip curves for pullout specimens with CFRP bars exposed to different temperatures.

6. Recommendations

Additional experimental work should be carried out to verify the proposed analytical model using other types of commercially available FRP bars.

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Appendix A

A.1. Sample of calculations

The analytical equation, which represents the ascending part of the CFRP bar's bond-slip curve after exposure to 125 °C, can be determined as follows. Parameters ($A = 0.564$, $B = 0.184$, $C = 0.186$, $D = 1.047$, $E = 0.582$, $F = 0.0$) were obtained from Table 8 and used to calculate the variables ϕ and β .

$$\begin{aligned}\phi_{125} &= 0.564 \times |0.672 - 0.184 - (5.285 \times 125 \times 10^{-4}) \\ &\quad - (2.983 \times 125^2 \times 10^{-6})|^{0.186} \\ &= 0.4700\end{aligned}$$

$$\beta_{125} = \left\{ 1.047 \times \left[0.507 - \left(\frac{125}{1314} \right)^{0.856} \right]^{0.582} \right\} + 0.0 = 0.5902$$

$$\text{For CFRP bars @ 125C} \quad \tau_{(s,125)} = 0.47 \times \tau_{m,23^\circ} \times (s)^{0.5902}$$

$$\begin{aligned}\text{For CFRP bars @ 125C} \quad s_{max,125} &= s_{max,23} \times \left(1 - \frac{125^{2.287}}{10^6} \right) \\ &= 0.9375 \times s_{max,23}\end{aligned}$$

The experimental pullout tests of CFRP bars at 23 °C showed that maximum bond strength $\tau_{m,23^\circ}$ was 8.338 MPa at corresponding slip $s_{max,23^\circ}$ of 3.278 mm.

$$\text{For CFRPs} s_{max,125} = 0.9375 \times 3.278 = 3.073 \text{ mm}$$

$$\tau_{(s,125)} = 0.470 \times 8.338 \times (s)^{0.5902} = 3.919 \times (s)^{0.5902}$$

$$\tau_{(max,125)} = 0.470 \times 8.338 \times (3.073)^{0.5902} = 7.6 \text{ MPa}$$

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